ARC-AGI-3 Symbolica

Reflexion-driven hypothesis-test agent on ft09 — what works, where the L4 ceiling sits

ARC-AGI-3 Symbolica Research

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Since v40 baseline — in one slide

[added] *since v40*

Reflexion module on stagnation
Abstract+grounded predicates
Multi-step plan with abort
Strict region_id binding

→

[worked]

Confirm rate **6%** → **52%**
L2–L3 reached in <30 turns
Cross-level transfer banked
Rationale coverage 73% → 96%

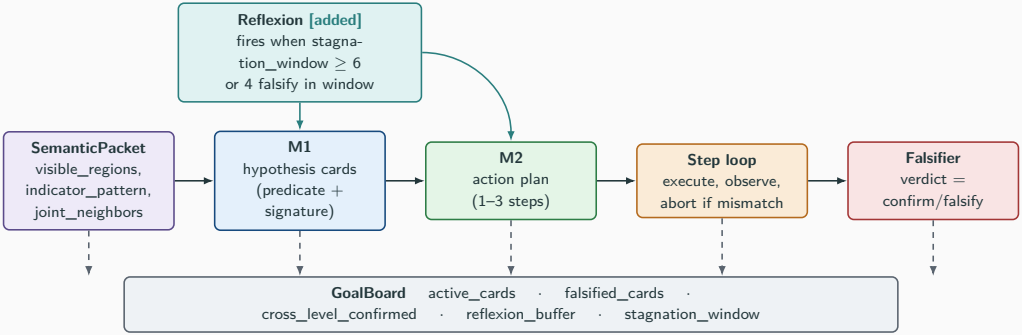
→

[missing]

L4 ceiling unbroken
0 NTi clicks despite 3 NTi
56% off-region at L4
Multi-step plan rarely emitted

v40 = palette tokens, no Reflexion. v41-c = abstract+grounded, Reflexion. v43 = discovery + anomaly hypothesis (running).

Per-turn loop — five components, one shared GoalBoard



M3 (skill compressor) and M4 (lesson) are gated off in v41+. Reflexion took over the corrective channel.

ft09 has six levels — L4 is where the mechanic changes

level	name	markers	Hkx	NTi	gqb cycle	kCv budget
L0	THR	1	8	0	{9, 8}	32
L1	hxv	1	13	0	{9, 12}	32
L2	Fmh	2	23	0	{8, 12}	96
L3	oea	2	18	0	{9, 8, 12}	96
L4	INW	6+	27	**3 NTi**	{14, 15}	128
L5	DFx	1	0	ZkU	—	128

L0–L3 share Hkx single-toggle physics with growing palette + count. L4 introduces **NTi** (multi-toggle). L5 introduces **ZkU** (custom).

Predicate abstraction — palette token cannot transfer L0→L4

(v40) Palette-only predicate — works only on L0:

```
predicate: "R31 transitions 9 to 8"
problem  : L1 cycle is {9,12}, L2 is {8,12}, L3 is {9,8,12}, L4 is {14,15}.
           "9 to 8" tokens have ZERO transfer value when palette rotates.
```

(v41-c) Abstract intent + concrete grounding — transfers across levels:

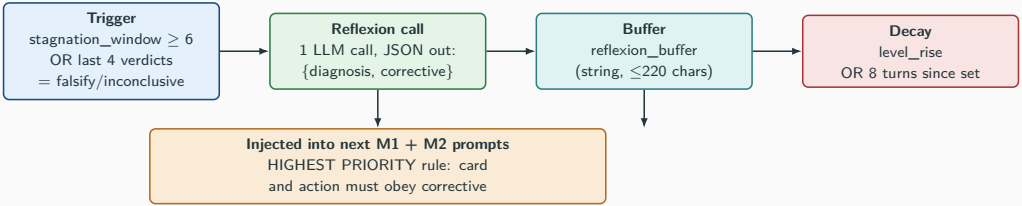
```
predicate: "Marker R31 reaches satisfied state when satisfy marker.E
            needs_toggle is completed by advance neighbor toward
            target_color (currently R32 9 -> 8)."
form     : <ABSTRACT INTENT>  <CONCRETE GROUNDING in parens>
abstract : "satisfy marker.<dir> needs_toggle", "advance neighbor toward
            target_color", "shared neighbor with conflicting target_colors"
grounding: "(currently R32 9 -> 8)"  <-- still satisfies BIND_DIGIT_RE.
```

Hidden bug fixed alongside the rewrite:

```
BIND_DIGIT_RE was:  r"r\d+[^.]{0,80}?\b\d\s*<conn>\s*\d"        # single digit only
                    -> silently rejected "14 to 15", "8 to 12" at L1+
fixed to:           r"r\d+[^.]{0,80}?\b\d{1,2}\s*<conn>\s*\d{1,2}"
```

Without the regex fix the abstraction would have been silently filtered out.

Reflexion — one LLM call per stagnation, $\leq$220-char corrective



Replaces v40 M3/M4 corrective channel which never reached the action planner. Reflexion writes directly to M1+M2.

Cross-level analogy bank — agent-confirmed predicates from cycle40

Real entries in `board.cross_level_confirmed` after L0→L3 progression:

```
[1] "Win-state for marker R18 is satisfy marker.NE needs_toggle by advance
     neighbor toward target_color (currently R15 12 -> 8), so the same R15
     transition jointly advances R18 and R10 toward completion."
     [precision_score 84, level_delta 1, is_target_state true]

[2] "Mask-rule alternative on marker R18: click neighbor advances one step
     in gqb cycle (currently R15 12 -> 9) before final target 8, i.e.,
     satisfy marker.NE needs_toggle may require two-step cycling."

[3] "Win-state for marker R18 is satisfy marker.NE needs_toggle by advance
     neighbor toward target_color (currently per_neighbor_target.NE neighbor
     R15 9 -> 8), while shared neighbor with conflicting target_colors at
     R6 (9 -> 8 vs stay 9) remains deferred."
     [precision_score 99, is_target_state true]
```

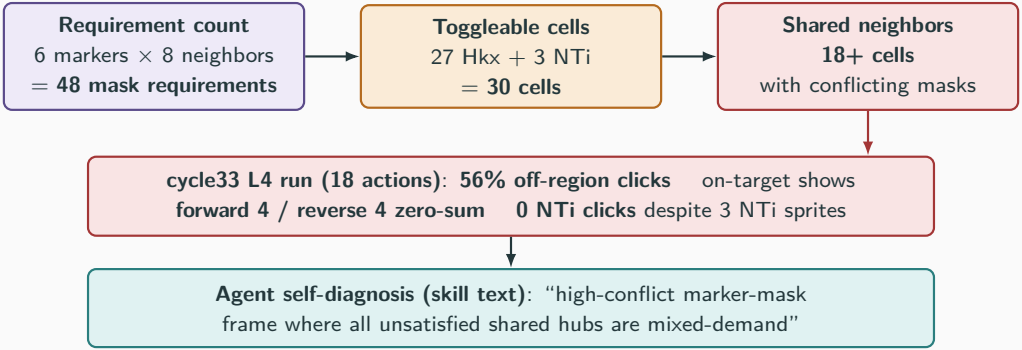
Abstract intent (*satisfy marker.<dir> needs_toggle*, *advance neighbor*) is constant. Only the parenthetical R-id and palette tokens change between levels.

Cycle progression — four cycles since palette baseline

cycle	rev	conf %	max L	multi-step	cross-L	remark
33	v40 (palette)	6%	L4 @ t33	0	0	5 redundant skills
37	v41-b (strict gates)	35%	L3 @ t53	0	2	slow climb
38	v41-c (abstract+grd)	**52%**	L2 @ t25	0	4	fastest L2 entry
40	v41-c rerun	50%	L3 @ t28	0	**6**	L3 stag 45, no L4
41	v43 (disc+anom)	—	—	—	—	running, L4-targeted

Confirm rate climbed 6% → 52% in three revs. Multi-step still 0 across all cycles. L4 ceiling unbroken.

L4 ceiling — the structural arithmetic of why v41-c stalls



48 demands vs 30 toggleables, >60% of toggles shared. Single-step planning cannot escape zero-sum hubs.

NTi mechanism — one click advances four Hkx at once

From `environment_files/ft09/9ab2447a/ft09.py` lines 2400–2424:

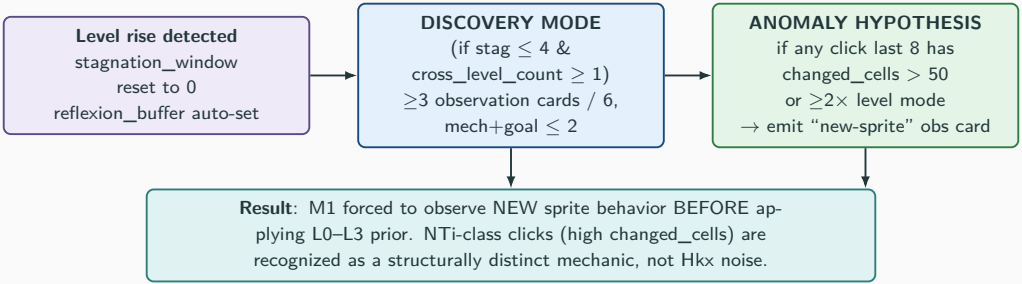
```
NTi sprite (introduced at L4):
  - placed alongside Hkx in the same play area
  - on click, NTi.pixel mask is consulted: every cell within the mask
    advances one step in the gqb cycle, simultaneously
  - effective behaviour: 1 click -> ~4 Hkx neighbours change in one step
```

Why M1 (v41-c) cannot see this:

```
- per_neighbor_target schema models 1-marker-per-click physics only.
- changed_cells statistic at L0-L3 is ~36 (one neighbor block).
- An NTi click would produce changed_cells ~ 4 x 36 = ~144, far above
  the L0-L3 mode. v41-c has no anomaly rule -- M1 silently treats the
  outlier as noise and re-emits Hkx-style cards.
```

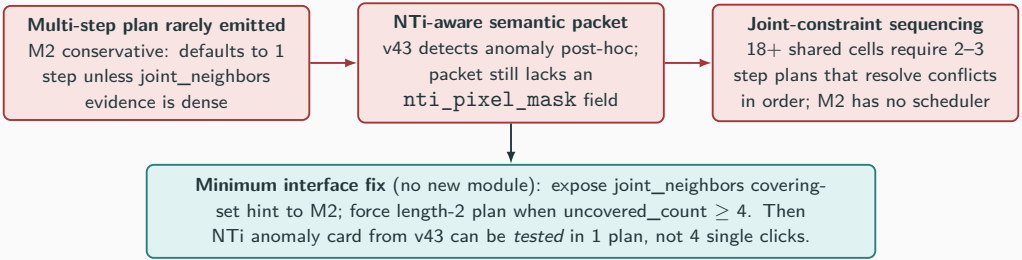
The mechanic is in the env; the agent's semantic packet does not surface it.

v43 fix — discovery mode and anomaly hypothesis on level rise



Currently running as cycle41. Targets the L4 ceiling identified on slide 9.

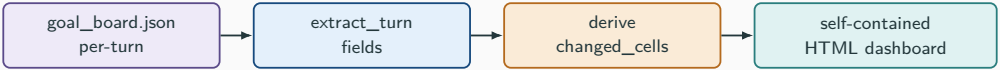
What is still missing for L4 clear — three gaps after v43



v43 fixes "M1 doesn't notice NTi". It does not fix "M2 cannot plan multi-click".

Visualizer — ft09_trace_explorer.html bundles five cycles

field	value
file size	3.7 MB self-contained HTML
cycles bundled	5 (cycle33, 37, 38, 40, 41)
per-turn fields surfaced	20
key per-turn diff fields	`reported_changed_cells` vs `derived_changed_cells`
rationale-coverage cycle33 (v40)	73%
rationale-coverage cycle40 (v41-c)	**96%**



File: `presentation/ft09_trace_explorer.html`. Coverage 73% → 96%.